Technology Stack

Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap, Jinja2 Templates	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results. Jinja2 enables dynamic rendering of prediction output.
2	**Backend / Server-Side**	Python 3, Flask	Flask handles HTTP requests, processes user input, communicates with the ML model, and returns prediction results. Python provides strong support for machine learning.
3	**Database / Data Storage**	CSV Dataset (credit_card_data.csv)	Stores historical applicant records used for model training. Lightweight and suitable for machine learning experiments and academic projects.
4	**Cloud / Hosting / Deployment**	Localhost (Flask Development Server) (Can also be deployed on Render or Heroku)	Used for testing and demonstrating the application. Easy deployment and maintenance with minimal configuration.
5	**Version Control & CI/CD**	Git & GitHub	Enables version control, collaborative development, code backup, and project management.
6	**Third-Party APIs / Other**	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib,	Scikit-learn builds and evaluates the prediction model. Pandas and

	Tools	Seaborn	NumPy handle data preprocessing. Joblib saves and loads the trained model. Matplotlib and Seaborn generate data visualizations and analysis charts.